INCH-POUND

MIL-D-52349C(ME)
22 August 1989
SUPERSEDING
MIL-D-52349B(ME)
21 November 1980

MILITARY SPECIFICATION

DISPLACEMENT AND EVACUATION KIT, HOSELINE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a displacement and evacuation kit consisting of ball inlet, ball receiver, displacement ball, grooved couplings, and pipe caps and items designed to remove liquid fuel and air from a hoseline by propelling the displacement ball through the hoseline under air pressure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

WW-C-633	- Coupling Halves, Cap and Wye; Quick Disconnect Pneumatic Hose, Two-Lug Universal Type.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3835

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MIL-C-10387	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe.
MIL-P-10388	- Pipe Fittings: One or More Ends Grooved.
MIL-H-52262	- Hose Assembly, Rubber: Lightweight, Collapsible, 4-Inch.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1472	- Human Engineering Design Criteria for Military System, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13200E6226	- Ball Inlet.
13200E6229	- Ball Receiver.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSH, Fort Belvoir, VA 22060-5606).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hoseline displacement and evacuation kit, hereinafter called "kit", shall consist of the following components fitted in a chest:

Component	Quantity
Ball inlet	1
Ball receiver	1
Displacement ball	1

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Grooved couplings	8
Pipe caps	16
Ejector assembly	1

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, and processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production items shall be made available upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.3.2).

3.3.1 Material deterioration prevention and control. The kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the kit may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the kit may be newly fabricated from recovered materials to the maximum extent practicable, provided the kit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the kit.

3.4 Ball inlet. The ball inlet shall be as shown on Drawing 13200E6226.

3.5 Ball receiver. The ball receiver shall be as shown on Drawing 13200E6229.

3.6 Displacement ball. The displacement ball shall be manufactured from polyurethane foam with a density not less than 2 lbs/ft^3 (32 kg/m^3), 4.50 $\pm$0.25 inches (114.3 mm $\pm$6.35 mm) outside diameter and 8.0 $\pm$0.5 inches (203.2 mm $\pm$12.7 mm) long.

3.7 Grooved couplings. The grooved couplings shall be in accordance with MIL-C-10387, class A, size 4.00 inch (101.6 mm).

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3.8 Pipe caps. The aluminum alloy pipe caps shall be in accordance with MIL-P-10388, size 4.00 inch (101.6 mm).

3.9 Ejector assembly. The ejector assembly shall be a Penberthy ejector, model GL-1, or equal, equipped with universal, pneumatic, quick-disconnect half couplings in accordance with WW-C-633, type III, connected to the inlet and suction side of the ejector. The ejector assembly shall be capable of creating a vacuum of not less than 20 inches (67.54 kPa) of mercury when supplied with a driving medium of compressed air at 80 pounds per square inch (psi) (552 kPa) with a volume of 110 cubic feet per minute (ft^3/min) (3.11m^3/min).

3.10 Performance. The kit shall displace water and evacuate air from 1000 feet (304.8 m) of 4.00-inch (101.6 mm), petroleum, oils and lubricants (pol), collapsible hoseline. Time for each operation shall not exceed 15 minutes.

3.10.1 Pressure relief. The ball receiver shall automatically relieve the air pressure after arrival of the ball.

3.11 Chest. A steel chest shall be provided for the components of the kit. The chest shall be not more than 22 inches (558.8 mm) in height, and shall be constructed of steel not less than 0.073 inch (1.85 mm) thick. The chest shall have a hinged cover and latches to secure the cover in the closed position. The cover shall be self-supporting in the open position. Handles shall be in accordance with MIL-STD-1472, 5.9.11.5. The chest cover and front shall be labeled "HOSELINE, DISPLACEMENT AND EVACUATION KIT" in one inch (25.4 mm) high letters. Bins, cradles, or blocking shall be provided for the components of the kit to prevent damage during shipment and handling. Break all exposed sharp edges and corners and deburr.

3.11.1 Loading plan. A loading plan indicating the location of all items in the chest shall be furnished. The loading plan shall be permanently attached to the inside of the cover.

3.12 Treatment and painting. The portion of the kit normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383.

3.13 Identification marking. The kit shall be identified in accordance with MIL-STD-130. Identification marking shall be applied to the outside of the cover of the chest.

3.14 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.5):

Item No.	Description	Identification	Quantity
1.	Hose Assembly, Rubber: Lightweight, Collapsible, 4 inch (101.6 mm)	MIL-H-52262	1000 ft. (304.8 m)

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3.15 Workmanship. All components of the kit and chest shall be free of harmful extraneous material such as sand, dirt, sprues, scale, and flux. External surfaces shall be free from burrs, sharp edges, and corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services are in accordance with prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. The sample unit for this inspection shall be one complete kit.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The kit shall be examined as specified herein for the defects listed in table I.

TABLE I. Examination schedule.

Number	Defects	Requirement paragraph
Major		
101.	Components not as specified.	3.1
102.	Dimensions and tolerances not as specified on drawings.	3.1.1
103.	First article not as specified.	3.2
104.	Material not as specified.	3.3
105.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
106.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
107.	Contractor does not have documentation available for identification of material and material finishes, or treatment.	3.3.1.2
108.	Used, rebuilt or remanufactured components, pieces, or parts are incorporated in the kit.	3.3.2
109.	Ball inlet not as shown on drawing 13200E6226.	3.4
110.	Ball receiver not as shown on drawing 13200E6229.	3.5
111.	Displacement ball not manufactured as specified.	3.6
112.	Grooved couplings not in accordance with MIL-C-10387.	3.7
113.	Pipe caps not in accordance with MIL-P-10388.	3.8
114.	Ejector assembly not as specified.	3.9
115.	Performance not as specified.	3.10
116.	Pressure relief not as specified.	3.10.1
117.	Chest not as specified.	3.11
118.	Loading plan missing.	3.11.1
119.	Components not packed in chest in accordance with loading plan.	3.11.1

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TABLE I. Examination schedule. (Cont'd)

Number	Defects	Requirement paragraph
120. Minor	Workmanship not as specified.	3.15
201.	Treatment and painting not as specified.	3.12
202.	Identification marking not in accordance with MIL-STD-130.	3.13

4.5.2 Tests.

4.5.2.1 Displacement. Remove the hose from the flaking box. Fill the hose with water. Insert the displacement ball in the hose and attach the ball inlet. Attach the ball receiver to the other end of the hose. Connect an air supply capable of producing a volume of 110 ft^3/min. (3.11 m^3/ min) at 80 psi (552 kPa) to the ball inlet. Supply air at approximately 100 ft^3/min (2.83 m^3 min). Continue supplying air until the displacement ball reaches the ball receiver. Inability of the displacement ball to displace the water from the hose within 15 minutes or to travel through the hose until it is caught in the ball receiver shall constitute failure of this test.

4.5.2.2 Evacuation. Remove the ball receiver and displacement ball, and cap the hoseline. Connect the ejector assembly to the ball inlet. Connect an air supply capable of producing a volume of 110 ft^3/min (3.11 m^3 min) at 80 psi (552 kPa) to the inlet of the ejector assembly. Inability to evacuate and collapse the 1,000 feet (304.8 m) of hoseline within 15 minutes shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Major

121. Land-sealing surface of grooved end fittings not completely covered with barrier material and tape as specified.
122. Application of barrier material and tape not in accordance with figure 1, as specified.
123. Shipping containers not as specified for level A or B.

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124. Chest lid not sealed with tape as specified.
125. Marking missing, illegible, incorrect or incomplete for level A, B, or C.

5. PACKAGING

5.1 Preservation (see 6.4). The land-sealing surface of grooved end fittings shall be completely covered with a single wrap of barrier material in accordance with MIL-B-121, type II, class 2 (see figure 1). The wrap for the land-sealing surface shall be flush with the end of the fitting and shall extend back beyond the groove not less than 0.250 inch (6.35 mm). The wrap of the barrier material shall be covered with a single wrap of tape in accordance with PPP-T-60, type IV. The tape shall be of a width that will completely cover the barrier material and shall extend not less than 0.125 inch (3.18 mm) beyond the end of the fitting and extend back beyond the barrier material not less than 0.500 inch (12.7 mm). The end of the tape extending beyond the end of the fitting, and the tape extending beyond the wrap shall be rolled in place against the end wall of the fitting and against the fitting (see figure 1). The components comprising the complete kit shall then be placed in the chest in accordance with the loading plan. After closure, all edges of the chest lid shall be sealed with tape in accordance with PPP-T-60, type IV.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The chest containing the kit components shall be packed in a close-fitting box in accordance with PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification, except the strapping shall be flat and finish B.

5.2.2 Level B. Packing shall be as specified for level A except the boxes shall be domestic type, and strapping may be finish A.

5.2.3 Level C. The chest containing the kit components shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use with the 4.00-inch (101.6 mm), lightweight, collapsible hoseline fuel system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Timeframe required for submission of first article and number of kits required (see 3.2).

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d. When the Government will conduct any or all of the first article examinations and tests. When the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests will be conducted by the contractor (see 3.2).
e. When the color of the kit should be other than as specified (see 3.12).
f. Level of packing required (see 5.2).
g. Any special marking (see 5.3).

6. First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Levels of protection. For marking purposes, the level of preservation and packing shall be designated A/A, A/B, or A/C.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.14.

6.6 Subject term (key word) listing.

Displacement and evacuation kit, hoseline
Kit, displacement and evacuation

6.7 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: 8/2/99).

6.8 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

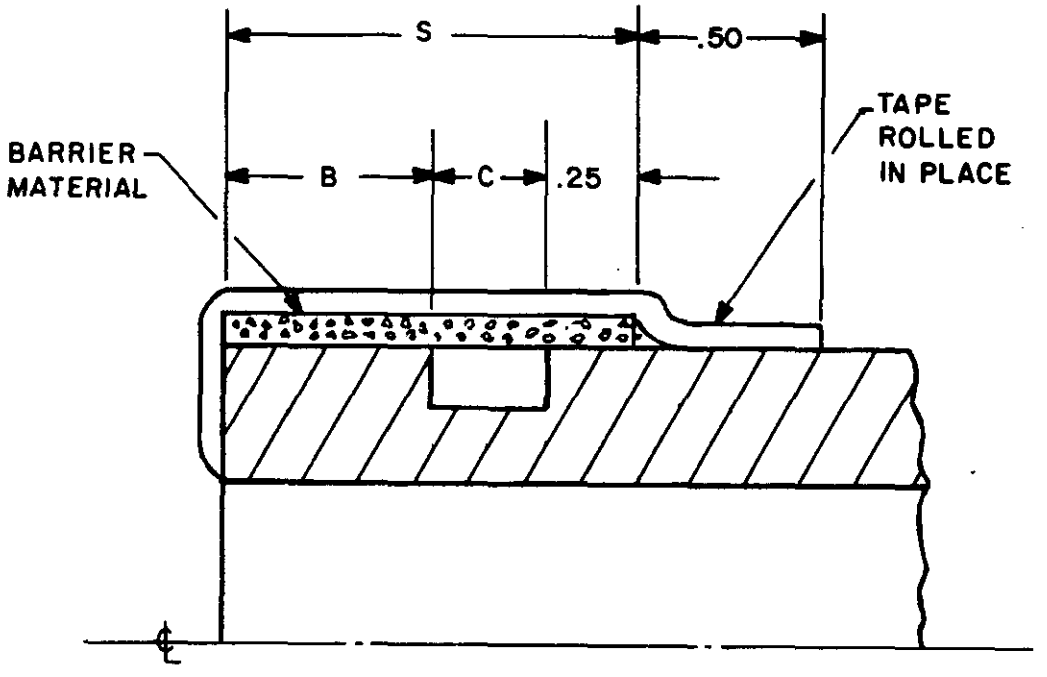
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S ——— LAND SEALING SURFACE

B ——— LAND

C ——— GROOVE

NOTE :

1. ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Land sealing surface for grooved end pipe, tube end fittings.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-52349C(ME)	Displacement and Evacuation Kit, Hoseline

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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